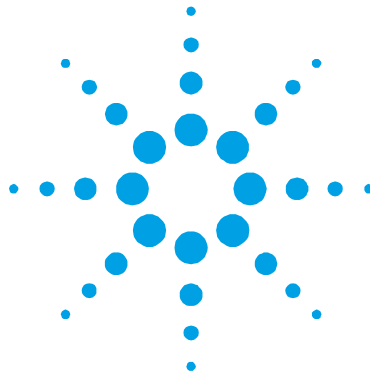




TwisTorr 305 FS onboard Controller

Models

X3507-64130

X3507-64131

Manuale di istruzioni

Bedienungshandbuch

Notice de mode d'emploi

Manual de instrucciones

用户手册

ユーザーマニュアル

User Manual

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04/2020



Agilent Technologies

Notices

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CAUTION

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WARNING

A **WARNING** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in personal injury or death. Do not proceed beyond a **WARNING** notice until the indicated conditions are fully understood and met.

TwisTorr 305 FS onboard Controller



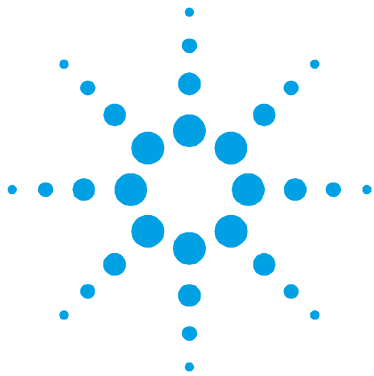
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Istruzioni per l'uso

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Traduzione delle istruzioni originali



Informazioni Generali

Questa apparecchiatura è destinata ad uso professionale.

L'utilizzatore deve leggere attentamente il presente manuale di istruzioni ed ogni altra informazione addizionale fornita dalla Agilent prima dell'utilizzo dell'apparecchiatura. La Agilent si ritiene sollevata da eventuali responsabilità dovute all'inosservanza totale o parziale delle istruzioni, ad uso improprio da parte di personale non addestrato, ad interventi non autorizzati o ad uso contrario alle normative nazionali specifiche.

Il controller TwisTorr 305 FS onboard è un controller adatto ad essere utilizzato con le pompe TwisTorr 305. Ha dei connettori ausiliari tramite i quali è possibile alimentare un ventilatore aggiuntivo, comandare la valvola di vent, pilotarlo da remoto tramite un computer host collegato con linea seriale (RS232 o RS485). Nei paragrafi seguenti sono riportate tutte le informazioni necessarie a garantire la sicurezza dell'operatore durante l'utilizzo dell'apparecchiatura. Informazioni dettagliate sono fornite nell'appendice "Technical information".

Questo manuale utilizza le seguenti convenzioni:

ATTENZIONE!

I messaggi di attenzione sono visualizzati prima di procedure che, se non osservate, potrebbero causare danni all'apparecchiatura.

AVVERTENZA!



I messaggi di avvertenza attirano l'attenzione dell'operatore su una procedura o una pratica specifica che, se non eseguita in modo corretto, potrebbe provocare gravi lesioni personali.

NOTA

Le note contengono informazioni importanti estrapolate dal testo.

Immagazzinamento

Durante il trasporto e l'immagazzinamento del TwisTorr 305 FS onboard non devono essere superate le seguenti condizioni ambientali:

- temperatura: da -20 °C a +70 °C
- umidità relativa: 0 – 95 % (non condensante)

Preparazione per l'installazione

Il TwisTorr 305 FS onboard viene fornito in un imballo protettivo speciale; se si presentano segni di danni, che potrebbero essersi verificati durante il trasporto, contattare l'ufficio vendite locale. Non disperdere l'imballo nell'ambiente. Il materiale è completamente riciclabile e risponde alla direttiva CEE 85/399 per la tutela dell'ambiente.

Installazione

Non installare e/o utilizzare il controller in ambienti esposti ad agenti atmosferici (pioggia, gelo, neve), polveri, gas aggressivi, in ambienti esplosivi o con elevato rischio di incendio. Durante il funzionamento è necessario che siano rispettate le seguenti condizioni ambientali:

- temperatura: da + 5 °C a +45 °C
- umidità relativa: 0 – 90 % (non condensante)

AVVERTENZA!



Il Turbo controller (nella sua versione a tensione di rete, X3507-64131) deve essere alimentato mediante un cavo di alimentazione a tre fili (vedere tabella delle parti ordinabili) con una spina di tipo approvato a livello internazionale ai fini della sicurezza dell'utente. Utilizzare sempre questo cavo di alimentazione ed inserire la spina in una presa con un adeguato collegamento di terra onde evitare scariche elettriche e per rispettare le specifiche CE. All'interno del controller si sviluppano alte tensioni che possono recare gravi danni o la morte. Prima di eseguire qualsiasi operazione di installazione o manutenzione del controller scollegarlo dalla presa di alimentazione.

AVVERTENZA!



Per evitare danni alle persone ed all'apparato, nel caso in cui la pompa sia appoggiata su di un tavolo assicurarsi che sia stabile. Non azionare mai la pompa Turbo se la flangia di ingresso non è collegata al sistema o non è chiusa con la flangia di chiusura.

ATTENZIONE!

Il TwisTorr 305 FS onboard appartiene alla seconda categoria di installazione (o sovratensione) prevista dalla normativa EN 61010-1. Connettere quindi il dispositivo ad una linea di alimentazione che soddisfi tale categoria.

Il TwisTorr 305 FS onboard ha dei connettori per gli ingressi/uscite e per la comunicazione seriale che devono essere connessi ai circuiti esterni in modo che nessuna parte sotto tensione sia accessibile.

Assicurarsi che l'isolamento del dispositivo connesso al TwisTorr 305 FS onboard abbia un isolamento adeguato anche in condizione di guasto singolo come previsto dalla normativa EN 61010-1.

Per l'installazione degli accessori opzionali, vedere "Technical Information".

Uso

In questo paragrafo sono riportate le principali procedure operative. Prima di azionare il Controller Turbo, eseguire le connessioni elettriche e collegare il manicotto di pre vuoto facendo riferimento al Manuale d'istruzioni.

Accensione ed Uso del TwisTorr 305 FS onboard

Per accendere il TwisTorr 305 FS onboard è sufficiente fornire la tensione di alimentazione. Il controller riconosce automaticamente la presenza dei segnali di interlock e di START ed avvia la pompa.

Alla prima accensione, il controller avvia automaticamente la pompa in modo "soft start". Attendere il completamento dell'avvio in soft start (quindi attendere che la pompa raggiunga la massima velocità di rotazione, indicata da LED verde acceso fisso), prima di spegnere la pompa.

L'avvio della pompa in modalità soft start permette la corretta redistribuzione del lubrificante all'interno dei cuscinetti, necessaria dopo un lungo periodo di inattività della pompa).

Il periodo di fermo dopo il quale è necessario procedere ad un nuovo avvio in Soft Start è di 60 giorni e dopo il primo avvio (per il quale il controller esegue il soft start automaticamente), le successive attivazioni dell'avvio in Soft Start sono a carico dell'utente.

Per avere una partenza con "Soft Start" attivo occorre abilitare il modo suddetto tramite software o agendo sul connettore ingressi e uscite remote (Remote I/O connector).

Per la procedura di attivazione della modalità "Soft Start" fare riferimento al paragrafo "Signal Description" nel capitolo "Technical Information". Il LED verde posto sul pannello del TwisTorr 305 FS onboard indica, con la frequenza del suo lampeggio, le condizioni operative del sistema:

- acceso fisso: la pompa è in rotazione normale;
- lampeggiante lentamente (periodo di circa 400 ms): il sistema è in stato di rampa, o di Stop, o di "Waiting for interlock";
- lampeggiante velocemente (periodo di circa 200 ms): condizione di errore.

Arresto del TwisTorr 305 FS onboard

Per arrestare il TwisTorr 305 FS onboard è sufficiente togliere la tensione di alimentazione.

Arresto di Emergenza

Per arrestare in condizioni di emergenza il TwisTorr 305 FS onboard occorre staccare il cavo di alimentazione.

Manutenzione

Il TwisTorr 305 FS onboard non richiede alcuna manutenzione. Qualsiasi intervento deve essere eseguito da personale autorizzato. Prima di effettuare qualsiasi intervento sul sistema scollegarlo dall'alimentazione. In caso di guasto, è possibile usufruire del servizio di riparazione Agilent o del servizio "Agilent advanced exchange service" che permette di ottenere un controller rigenerato in sostituzione di quello guasto.

AVVERTENZA!



Prima di eseguire qualsiasi intervento sul controller, scollegarlo dall'alimentazione.

NOTA

Prima di rispedire al costruttore un sistema per riparazioni o advanced exchange service, è indispensabile compilare e far pervenire al locale ufficio vendite la scheda "Richiesta di ritorno" allegata al presente manuale di istruzioni. Copia della stessa deve essere inserita nell'imballo del sistema prima della spedizione.

Qualora un sistema dovesse essere rottamato, procedere alla sua eliminazione nel rispetto delle normative nazionali specifiche.

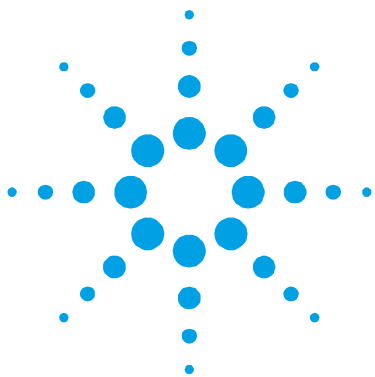
Smaltimento

Significato del logo "WEEE" presente sulle etichette. Il simbolo qui sotto riportato è applicato in ottemperanza alla direttiva CE denominata "WEEE". Questo simbolo (**valido solo per i paesi della Comunità Europea**) indica che il prodotto sul quale è applicato, NON deve essere smaltito insieme ai comuni rifiuti domestici o industriali, ma deve essere avviato ad un sistema di raccolta differenziata. Si invita pertanto l'utente finale a contattare il fornitore del dispositivo, sia esso la casa madre o un rivenditore, per avviare il processo di raccolta e smaltimento, dopo opportuna verifica dei termini e condizioni contrattuali di vendita.



Per maggiori informazioni riferirsi a:

<http://www.agilent.com/environment/product/index.shtml>



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Übersetzung der Originalanleitung



Allgemeine Informationen

Diese Vorrichtung ist für eine professionelle Anwendung bestimmt. Der Benutzer muss, vor Anwendung, vorliegendes Handbuch und alle weitere von Agilent gelieferte Angaben, aufmerksam durchlesen. Agilent ist für etwaige auf teilweise oder gesamte Nichtberücksichtigung der Gebrauchsanweisungen beruhende Verantwortungen, für eine nicht geeignete Anwendung durch nicht ausgebildetes Personal, für nicht autorisierte Eingriffe oder für Anwendung unter Nichtbeachtung der nationalen Bestimmungen, nicht verantwortlich. Der Controller TwisTorr 305 FS onboard ist für eine Anwendung mit Pumpen des Typs TwisTorr 305, geeignet. Er besitzt Hilfsverbinder, durch die ein zusätzlicher Ventilator gespeist und das Lüftungsventil betrieben werden können, eine Fernsteuerung ist durch einen, mit einer seriellen Linie verbundenen Host-Rechner, möglich (RS232 oder RS485). In den folgenden Absätzen sind alle notwendigen Informationen über die Sicherheit des Bedienungspersonals, während des Betriebs, angegeben. Ausführliche Angaben sind im Anhang "Technical Information", enthalten.

Dieses Handbuch benutzt folgende konventionelle Angaben:

VORSICHT!

Die Vorsichtshinweise erscheinen vor Verfahren, die bei Nichtbeachten, Geräteschaden verursachen könnten.

WARNUNG!



Die Warnhinweise deuten auf ein Verfahren oder ein besonderes Verhalten hin, das bei Nichtbeachtung der Vorsichtsmassnahmen, schwere persönliche Schäden verursachen könnte.

HINWEIS

Die Hinweise enthalten wichtige Informationen aus dem Text.

Lagerung

Während des Transportes und der Lagerung des TwisTorr 305 FS onboard, dürfen folgende Umgebungsbedingungen nicht überschritten werden:

- Temperatur: von -20 °C bis +70 °C
- relative Luftfeuchtigkeit: 0 – 95 % (ohne Kondensbildung)

Vorbereitung für die Installation

Der TwisTorr 305 FS onboard wird in einer speziellen Schutzverpackung geliefert; sollten während des Transports Schäden aufgetreten sein, verständigen Sie bitte die lokale Verkaufsabteilung. Die Verpackung ordnungsgemäß entsorgen. Das Material ist vollständig recyclebar.

Installation

Den Controller nicht in Räumen die Regen, Frost, Schnee, Staub oder aggressivem Gas ausgesetzt sind oder in Räumen mit Explosions- oder hoher Brandgefahr, verwenden.

Während der Betätigung müssen folgende Umgebungsbedingungen berücksichtigt werden:

- Temperatur: von 5 °C bis +45 °C
- relative Luftfeuchtigkeit: 0 - 90% (ohne Kondensbildung)

WARNUNG!



Der Turbo Controller (in seiner Netzspannungsversion, X3507-64131) muss mit einem dreiadrigen Netzkabel (siehe Bestelltabelle) mit international zur Sicherheit des Benutzers zugelassenem Stecker angeschlossen werden. Es sollte immer dieses Netzkabel benutzt werden, das an eine korrekt geerdete Steckdose anzuschließen ist, um den CE Richtlinien zu entsprechen und Stromschläge zu vermeiden. Im Inneren des Controllers entstehen hohe Spannungen, die schwere Verletzungen verursachen und lebensgefährlich sein können. Vor jedem Montage- bzw. Wartungseingriff muss deshalb der Netzstecker gezogen werden.

WARNUNG!



Falls Sie die Pumpe auf einem Tisch ablegen, stellen Sie zuvor dessen Stabilität sicher, um Personen- und Geräteschäden zu vermeiden. Schalten Sie die Turbo Pumpe nie ein, wenn der Eingangsflansch nicht am System angeschlossen oder nicht mit dem Abschlussflansch geschlossen ist.

VORSICHT!

TwisTorr 305 FS onboard wird in die zweite Installationskategorie (Überspannung) der Norm EN 61010-1, eingestuft. Die Vorrichtung muss an eine Netzlinie angeschlossen werden, die dieser Kategorie entspricht. TwisTorr 305 FS onboard verfügt außer den, für die serielle Kommunikation vorgesehenen Konnektoren, auch Ausgangs- und Eingangskonnektore, die mit den äußeren Schaltungen verbunden werden müssen, sodass kein Teil unter Spannung zugänglich sein kann. Sicherstellen, dass die Isolierung der mit dem TwisTorr 305 FS onboard verbundenen Vorrichtung, auch im Einzelschadenfall, so wie von Norm EN 61010-1 vorgesehen, eine angebrachte Isolierung besitzt.

Für die Installation der Options-Nebeneinrichtungen, beachten Sie bitte den Abschnitt "Technical Information".

Anwendung

In diesem Absatz werden die wichtigsten Funktionsverfahren angegeben. Bevor Sie den Controller Turbo aktivieren, schließen Sie die Elektroanschlüsse an und schließen Sie die freie Muffe an, wie im Benutzerhandbuch beschrieben.

Anschalten und Betrieb des TwisTorr 305 FS onboard

Die elektrische Versorgung schaltet den TwisTorr 305 FS onboard automatisch an. Der Controller erkennt automatisch die Interlock- und Anschaltsignale und schaltet die Pumpe an.

Beim ersten Einschalten startet der Regler die Pumpe automatisch im "Softstart"-Modus. Warten Sie, bis der Softstart abgeschlossen ist (dann warten Sie, bis die Pumpe die maximale Drehzahl erreicht hat, die durch eine ständige grüne LED angezeigt wird), bevor Sie die Pumpe ausschalten.

Das Starten der Pumpe im Sanftanlauf-Modus ermöglicht die korrekte Umverteilung des Schmiermittels in den Lagern, die nach einer langen Zeit der Pumpeninaktivität erforderlich ist).

Die Stillstandszeit, nach der ein neuer Softstart erforderlich ist, beträgt 60 Tage, und nach dem ersten Start (für den die Steuerung den Softstart automatisch durchführt) ist der Benutzer verantwortlich für die nachfolgenden Softstart-Aktivierungen.

Um einen Start mit aktivem „Soft Start“ zu erzielen, ist es notwendig, diesen Modus durch die Software oder durch den Anschluss der Remote-Ein- und Ausgänge zu aktivieren) (Remote I/O connector).

Um den Modus "Soft Start" einzustellen, bitte den Absatz "Signal Description", im Kapitel "Technical Information", nachschlagen. Die grüne leuchtendende Diode auf der Schalttafel des TwisTorr 305 FS onboard, gibt mit ihrer Leuchtfrequenz den System-Funktionszustand an:

- eingeschaltet, nicht intermittierend, die Pumpe hat eine normale Drehung;
- langsames Aufleuchten (ca. 400 ms), das System befindet sich im Ausgangs- oder Stop-Stadium oder im Stadium "Waiting for interlock";
- schnelles Aufleuchten (ca. 200 ms), gibt einen Fehler an.

Abschalten des TwisTorr 305 FS onboard

Um den TwisTorr 305 FS onboard abzuschalten, genügt es die elektrische Versorgung abzuschalten.

Notabschaltung

Um den TwisTorr 305 FS onboard im Notfall abzuschalten, die elektrische Verbindung unterbrechen.

Wartung

Für den TwisTorr 305 FS onboard ist keine Wartung erforderlich. Jeder Eingriff unterliegt autorisiertem Personal. Vor jedem Eingriff, die elektrische Versorgung unterbrechen. Bei einem Störfall kann der Reparaturservice Agilent oder der Service "Agilent Advanced Exchange service" in Anspruch genommen werden, durch den man einen regenerierten Controller als Ersatz für den defekten erhält.

WARNUNG!

Trennen Sie den Controller vor jedem beliebigen Eingriff vom Netz.



HINWEIS

Bevor Sie dem Hersteller oder dem Advanced Exchange Service ein System zur Reparatur zurücksenden, muss die Karte "Sicherheit und Gesundheit", die diesem Handbuch beiliegt, ausgefüllt und dem lokalen Verkaufsbüro zugesandt werden. Eine Kopie davon muss der Systemverpackung beigelegt werden.

Im Falle einer Verschrottung des Systems, muss diese nach den nationalen Vorschriften erfolgen.

Entsorgung

Bedeutung des "WEEE" Logos auf den Etiketten. Das folgende Symbol ist in Übereinstimmung mit der EU-Richtlinie WEEE (Waste Electrical and Electronic Equipment) angebracht.

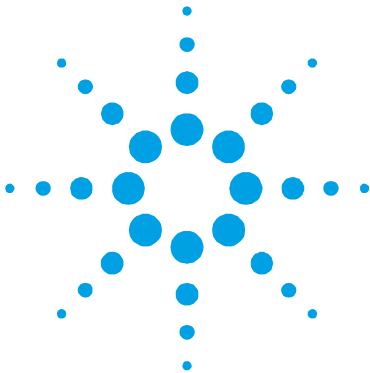
Dieses Symbol (**nur in den EU-Ländern gültig**) zeigt an, dass das betreffende Produkt nicht zusammen mit Haushaltsmüll entsorgt werden darf sondern einem speziellen Sammelsystem zugeführt werden muss.

Der Endabnehmer sollte daher den Lieferanten des Geräts - d.h. die Muttergesellschaft oder den Wiederverkäufer - kontaktieren, um den Entsorgungsprozess zu starten, nachdem er die Verkaufsbedingungen geprüft hat.



Weitere Informationen finden Sie unter:

<http://www.agilent.com/environment/product/index.shtml>



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Traduction de la mode d'emploi originale



Indications Generales

Cet appareillage a été conçu en vue d'une utilisation professionnelle. Avant toute utilisation de l'appareil, il est conseillé à l'utilisateur de lire attentivement cette notice d'instructions ainsi que toute autre indication supplémentaire fournie par Agilent qui décline par conséquent toute responsabilité en cas de non respect total ou partiel des instructions données, d'utilisation impropre par un personnel non formé, d'opérations non autorisées ou d'emploi contraire aux réglementations nationales spécifiques. Le TwisTorr 305 FS onboard est un contrôleur approprié pour être utilisé avec les pompes TwisTorr 305. Le TwisTorr 305 FS onboard est en outre doté de connecteurs auxiliaires permettant d'alimenter un ventilateur supplémentaire, de commander la vanne de ventilation, de le piloter à distance à l'aide d'un ordinateur host connecté par ligne série. Les paragraphes suivants fournissent toutes les indications nécessaires à garantir la sécurité de l'opérateur pendant l'utilisation de l'appareillage. Des renseignements plus détaillés se trouvent dans l'appendice "Technical Information".

Cette notice utilise les signes conventionnels suivants:

ATTENTION!

Les messages d'attention apparaissent avant certaines procédures dont le non respect pourrait endommager sérieusement l'appareillage.

AVERTISSEMENT!



Les messages d'avertissement attirent l'attention de l'opérateur sur une procédure ou une manoeuvre spéciale qui, effectuée de façon impropre, risque de provoquer de graves lésions.

NOTE

Les notes contiennent des renseignements importants, extrapolés du texte.

Stockage

Pendant le transport et le stockage du TwisTorr 305 FS onboard, veiller au respect des conditions environnementales suivantes:

- température: de -20 °C - à $+70\text{ °C}$
- humidité relative: 0 - 95 % (non condensante)

Préparation pour l'installation

Le TwisTorr 305 FS onboard est livré dans un emballage de protection spécial; en cas d'endommagement de l'emballage pouvant s'être produit pendant le transport, contacter le bureau de vente local. Ne pas abandonner l'emballage dans la nature. Le matériel est entièrement recyclable et conforme à la directive CEE 85/399 en matière de protection de l'environnement.

Installation

Ne pas installer et/ou utiliser le système dans des milieux exposés aux agents atmosphériques (pluie, froid, neige), poudre, gaz agressifs, dans des milieux explosifs ou avec risque élevé d'incendie. Pendant le fonctionnement, il est nécessaire de respecter les conditions environnementales suivantes:

- température: de +5 °C à +45 °C
- humidité relative: 0 - 90% (non condensante)

AVERTISSEMENT!



Le Turbo contrôleur (dans sa version avec tension secteur, X3507-64131) doit être alimenté au moyen d'un câble d'alimentation à trois fils (voir tableau des pièces que l'on peut commander) avec une fiche du type approuvé au niveau international en vue de la sécurité de l'utilisateur. Utiliser toujours ce câble d'alimentation et introduire la fiche dans une prise pourvue d'un branchement au sol approprié à la masse, afin d'éviter toute décharge électrique et de respecter les spécifications CE. A l'intérieur du contrôleur se développent de hautes tensions qui peuvent causer de graves dommages, voire la mort. Avant d'effectuer toute opération d'installation ou d'entretien du contrôleur, le débrancher de la prise d'alimentation.

AVERTISSEMENT!



Afin d'éviter des dommages aux personnes et à l'appareil, au cas où la pompe est appuyée sur une table, en vérifier la stabilité. Ne jamais actionner la pompe Turbo si la bride en entrée n'est pas branchée au système ou n'est pas fermée par une bride de fermeture.

ATTENTION!

Le TwisTorr 305 FS onboard appartient à la deuxième catégorie d'installations (ou surtension) prévue par la norme EN 61010-1. De ce fait, brancher le dispositif à une ligne d'alimentation compatible avec cette catégorie. Le TwisTorr 305 FS onboard dispose de connecteurs pour les entrées/sorties et pour la communication en série qui doivent être branchés aux circuits extérieurs de façon qu'aucune partie sous tension ne soit accessible.

S'assurer que l'isolation du dispositif branché au TwisTorr 305 FS onboard a une isolation appropriée même en condition de panne individuelle selon les termes de la norme EN 61010-1.

Pour l'installation des accessoires en option, se reporter à "Technical Information".

Utilisation

Ce paragraphe présente les principales procédures opérationnelles. Avant d'actionner le contrôleur Turbo, procéder aux branchements électriques et brancher le manchon de prévide en se référant au Manuel d'instructions.

Mise en marche et utilisation du TwisTorr 305 FS onboard

Pour allumer le TwisTorr 305 FS onboard il suffit de fournir la tension d'alimentation. Le contrôleur reconnaît automatiquement la présence de signaux d'interlock et de démarrage et il actionne la pompe.

Lors de la première mise en marche, le contrôleur démarre automatiquement la pompe en modalité « Soft Start ». Attendez que le démarrage progressif soit terminé (puis attendez que la pompe atteigne la vitesse de rotation maximale, indiquée par une DEL verte fixe), avant d'arrêter la pompe.

Le démarrage de la pompe en modalité « Soft Start » permet une redistribution correcte du lubrifiant à l'intérieur des paliers, ce qui est nécessaire après une longue période d'inactivité de la pompe).

La période d'arrêt après laquelle un nouveau démarrage Soft start est nécessaire est de 60 jours et après le premier démarrage (pour lequel le contrôleur effectue automatiquement le démarrage soft start), les activations ultérieures du démarrage soft start sont à la charge de l'utilisateur.

Pour avoir un démarrage avec « Soft Start » actif, il est nécessaire d'activer cette modalité par logiciel ou en agissant sur le connecteur des entrées et sorties à distance. (Remote I/O connector).

Pour la procédure d'activation de la modalité « Soft Start », se référer au paragraphe « SIGNAL DESCRIPTION" dans le chapitre "Technical Information)". La LED verte LD1 placée sur le panneau de la base du TwisTorr 305 FS onboard indique, par sa fréquence de clignotement, les conditions opérationnelles du système:

- allumée fixe: la pompe est en rotation normale;
- clignote lentement (période d'environ 400 ms): le système est en état de rampe, ou de stop, ou d'attente interlock (waiting for interlock).
- clignote rapidement (période d'environ 200 ms): condition d'erreur.

Arrêt du TwisTorr 305 FS onboard

Pour arrêter le TwisTorr 305 FS onboard, il suffit de retirer la tension d'alimentation. Le contrôleur arrête immédiatement la pompe.

Arrêt d'urgence

Pour arrêter le TwisTorr 305 FS onboard en conditions d'urgence, il faut débrancher le cordon d'alimentation.

Entretien

Le TwisTorr 305 FS onboard n'exige aucun entretien particulier. Toute intervention doit être effectuée par un personnel agréé. En cas de panne, on peut s'adresser au service de réparation Agilent ou au service "Agilent advanced exchange service" qui permet d'obtenir un contrôleur régénéré en substitution de celui endommagé.

AVERTISSEMENT!

Avant d'effectuer toute opération sur le contrôleur, le débrancher.



NOTE

Avant de renvoyer une pompe au constructeur pour réparation ou "advanced exchange service", remplir et faire parvenir au bureau Agilent de votre région la fiche "Instruction pour le retour du matériel" jointe au présent manuel d'instructions. Une copie de cette fiche devra être mise dans l'emballage de la pompe avant l'expédition.

En cas de mise au rebut de la pompe, procéder à son élimination conformément aux réglementations nationales concernant la gestion des déchets.

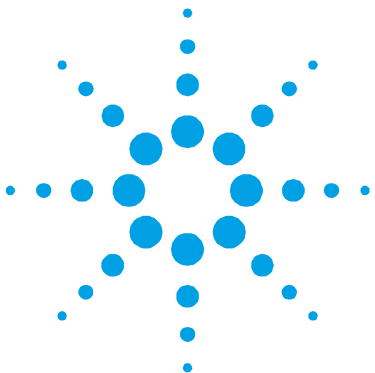
Mise au rebut

Signification du logo "WEEE" figurant sur les étiquettes. Le symbole ci-dessous est appliqué conformément à la directive CE nommée "WEEE". Ce symbole (**uniquement valide pour les pays de la Communauté européenne**) indique que le produit sur lequel il est appliqué NE doit PAS être mis au rebut avec les ordures ménagères ou les déchets industriels ordinaires, mais passer par un système de collecte sélective. Après avoir vérifié les termes et conditions du contrat de vente, l'utilisateur final est donc prié de contacter le fournisseur du dispositif, maison mère ou revendeur, pour mettre en œuvre le processus de collecte et mise au rebut.



Pour en savoir plus, consulter :

<http://www.agilent.com/environment/product/index.shtml>



4

Manual de instrucciones

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Traducción de las instrucciones originales



Informaciones Generales

El presente equipo está destinado a uso profesional. Antes de utilizar este sistema se aconseja al usuario que lea atentamente el manual de instrucciones así como cualquier otro tipo de información adicional que Agilent facilite.

Agilent se retiene libre de eventuales responsabilidades debido a la inobservancia total o parcial de las instrucciones, a la utilización incorrecta por parte de personal no adiestrado, a intervenciones no autorizadas o a un uso contrario con las normas nacionales específicas.

El controler TwisTorr 305 FS onboard es un controler apto para ser utilizado con las bombas TwisTorr 305. Está dotado de conectores adicionales mediante los que es posible alimentar un ventilador auxiliar, dirigir la válvula de vent y pilotarlo desde control remoto mediante un ordenador host conectado con una línea serial (RS232 ó RS485). En los siguientes párrafos se han incluido todas las informaciones necesarias para garantizar la seguridad del operador durante la utilización del equipo.

Para más información consultar el apéndice "Technical information".

Este manual utiliza las siguientes convenciones:

¡ATENCIÓN!

Los mensajes de atención aparecen cuando se está por realizar un determinado procedimiento que, en caso de no ejecutarse correctamente podría ser causa de daños en el equipo.

¡ADVERTENCIA!



Los mensajes de advertencia señalan al operador que un determinado procedimiento o una operación específica puede ocasionar graves lesiones personales sino se realizan de forma correcta.

NOTA

Las notas contienen informaciones importantes extrapoladas del texto.

Almacenamiento

Durante el transporte y el almacenamiento del TwisTorr 305 FS onboard no pueden ser superadas las siguientes condiciones ambientales:

- temperatura: de -20 °C a +70 °C
- humedad relativa: 0 – 95 % (no condensante)

Preparación para la instalación

El TwisTorr 305 FS onboard se suministra con un embalaje especial de protección; si el embalaje muestra signos de deterioro que podrían haberse verificado durante el transporte, será necesario contactar con el ente de ventas local. No dispersar el embalaje en el ambiente. El material es completamente reciclable y responde a la directiva CEE 85/399 sobre la tutela del ambiente.

Instalación

No instalar y/o utilizar el controler en ambientes expuestos a agentes atmosféricos (lluvia, hielo, nieve), polvo, gases agresivos, o en ambientes con riesgo elevado de explosión o de incendio. Durante el funcionamiento del controler es necesario que sean respetadas las siguientes condiciones ambientales:

- temperatura: de + 5 °C a +45 °C
- humedad relativa: 0 – 90 % (no condensante)

¡ADVERTENCIA!



El Turbo controler (en su versión con tensión de red, X3507-64131) debe ser alimentado mediante un cable de tres hilos (ver tabla de los componentes que pueden ser ordenados) con una clavija del tipo aprobado a nivel internacional con la finalidad de respetar las normas correspondientes a la seguridad del usuario. Utilizar siempre este cable de alimentación e introducir la clavija en un enchufe con una conexión de masa adecuada para evitar descargas eléctricas. y respetar las especificaciones CE. Dentro del controler se desarrollan altas tensiones que pueden causar graves daños o la muerte. Antes de efectuar cualquier operación de instalación o mantenimiento del controler desconectarlo del enchufe de alimentación.

¡ADVERTENCIA!



Siempre que la bomba se instale en una mesa, se aconseja comprobar que ésta sea estable para evitar daños a las personas y al aparato. No accione nunca la bomba Turbo si la abrazadera de entrada no está conectada al sistema o no está cerrada con la abrazadera de cierre.

¡ATENCIÓN!

El TwisTorr 305 FS onboard pertenece a la segunda categoría de instalación (o sobretensión) prevista por la normativa EN 61010-1. Conectar, por lo tanto, el dispositivo a una línea de alimentación que cumpla con los requisitos de dicha categoría. El TwisTorr 305 FS onboard está dotado de conectores para entradas/salidas y para la comunicación que deben ser conectados a los circuitos externos de forma que ninguna parte bajo tensión sea accesible. Comprobar que el aislamiento del dispositivo conectado al TwisTorr 305 FS onboard sea el adecuado incluso ante condiciones de avería individual tal como previsto en la normativa EN 61010-1.

Para la instalación de los accesorios opcionales, véase "Technical Information".

Uso

En este párrafo se han incluido los principales procedimientos operativos. Antes de accionar el TwisTorr 305 FS onboard, realice las conexiones eléctricas y conecte el manguito de pre-vacío siguiendo las instrucciones del Manual de Instrucciones.

Encendido y Utilización del TwisTorr 305 FS onboard

Para encender el TwisTorr 305 FS onboard es suficiente suministrar la tensión de alimentación. El controler reconoce automática-mente la presencia de las señales de interlock y de puesta en marcha y activa la bomba.

Al primer encendido, el controlador arranca automáticamente la bomba en modo "soft start". Espere a que se complete el arranque en soft start (luego espere a que la bomba alcance la velocidad máxima de rotación, indicada por un LED verde fijo), antes de apagar la bomba.

El arranque de la bomba en modo soft start permite la correcta redistribución del lubricante dentro de los rodamientos (lo cual es necesario después de un largo período de inactividad de la bomba).

Después de 60 días de inactividad es necesario proceder a una nueva puesta en marcha en Soft Start y después del primer arranque (para el cual el controlador efectúa el soft start automáticamente), las posteriores activaciones de puesta en marcha en Soft Start son a cargo del usuario.

Para arrancar con "Soft Start" activo, es necesario habilitar la modalidad mencionada a través del software o actuando sobre el conector de entradas y salidas remotas (Remote I/O connector).

Para el procedimiento de activación de la modalidad "Soft Start" consultar el punto "Signal Description" del capítulo "Technical Information". El LED verde ubicado en el panel del TwisTorr 305 FS onboard indica, con la frecuencia de su centelleo, las condiciones operativas del sistema:

- luz fija: la bomba está en rotación normal;
- centelleo lento (periodo de unos 400 ms): el sistema está en estado de rampa, o de Stop, o de "Waiting for interlock";
- centelleo rápido (periodo de unos 200 ms): condición de error.

Parada TwisTorr 305 FS onboard

Para parar el TwisTorr 305 FS onboard es suficiente quitar la tensión de alimentación.

Parada de Emergencia

Para parar el TwisTorr 305 FS onboard en condiciones de emergencia es necesario desconectar el cable de alimentación.

Mantenimiento

El TwisTorr 305 FS onboard no necesita ningún tipo de mantenimiento. Cualquier tipo de intervención sobre el sistema deberá ser realizado por personal autorizado. Antes de efectuar cualquier intervención sobre el sistema, es necesario desconectarlo de la alimentación. En caso de avería, es posible ponerse en contacto con el servicio de reparación Agilent o el servicio "Agilent advance exchange service" que permite obtener un controlador regenerado en sustitución del averiado.

¡ADVERTENCIA!



Antes de realizar cualquier intervención en el controlador, desconecte la alimentación.

NOTA

Antes de expedir al fabricante un sistema para su reparación o advanced exchange service, es indispensable rellenar y enviar a la oficina de ventas la ficha "Seguridad y salud " que se adjunta al presente manual de instrucciones. Una copia de dicha ficha deberá incluirse en el embalaje del sistema antes de la expedición.

En caso de que un sistema debiera ser desguazado, proceder a su eliminación respetando las normas nacionales específicas.

Eliminación

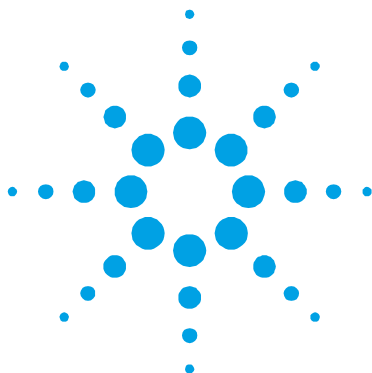
Significado del logotipo "WEEE" presente en las etiquetas.

El símbolo que se indica a continuación, es aplicado en observancia de la directiva CE denominada "WEEE". Este símbolo (**válido sólo para los países miembros de la Comunidad Europea**) indica que el producto sobre el cual ha sido aplicado, NO debe ser eliminado junto con los residuos comunes sean éstos domésticos o industriales, y que, por el contrario, deberá ser sometido a un procedimiento de recogida diferenciada. Por lo tanto, se invita al usuario final, a ponerse en contacto con el proveedor del dispositivo, tanto si éste es la casa fabricante o un distribuidor, para poder proveer a la recogida y eliminación del producto, después de haber efectuado una verificación de los términos y condiciones contractuales de venta.



Para obtener más información, consulte:

<http://www.agilent.com/environment/product/index.shtml>



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原始说明手册的译本



一般说明

如果在使用时不遵守或不完全遵守这些说明、未经培训的人员进行不正确使用、未经授权进行设备维修或在不符合国家特定标准进行使用，安捷伦不承担任何责任。

TwisTorr 305 FS板载控制器适用于TwisTorr 305泵。它配备辅助连接器，该辅助连接器可为附加风扇供电、控制通风阀以及通过串行线路（RS232或RS485）上连接的主机对TwisTorr 305 FS板载控制器进行远程操作。以下各段包含使用设备时确保操作员安全所需的所有信息。

有关详细信息，请参阅“技术信息”附录。

本手册使用以下标准协议：

小心!

小心消息会在操作过程之前显示，如果未注意到，可能会损坏设备。

警告!



警告消息让操作员特别注意特定的程序或操作，如果执行不当，可能导致严重的人身伤害。

注意

注意包含从文本中摘录的重要信息。

储存

运输和存储泵时，应满足以下环境要求：

- 温度： -20°C 至 $+70^{\circ}\text{C}$
- 相对湿度： 0 -95 % （无冷凝）

安装准备

TwisTorr 305 FS板载控制器拥有特殊的保护性包装； 如果在运输过程中出现损坏，请与当地销售中心联系。 请勿将包装物丢弃在环境中。 该材料完全可回收，且符合85/399/EE环保指令的规定。

安装

请勿在暴露于大气介质（雨、雪、冰）、灰尘、腐蚀性气体的环境中、爆炸性环境中或具有高火灾风险的环境中安装或使用控制器。在运行过程中，必须遵守以下环境条件：

- 温度：从+5° C至+45° C
- 相对湿度： 0 - 90 % （无冷凝）

警告!



为了用户安全，涡轮控制器（在带有电源电压的版本中，**X3507-64131**）必须由 **3** 线电源线（请参阅可订购的零件表）和插头（国际认可）供电。始终使用此电源线并将插头插入具有接地功能的适配插座中，以避免电击并符合 **EC** 要求。控制器内部产生的高压会可能导致严重的人身伤害甚至死亡。在安装或维修本机之前，请务必将其与插座断开。

警告!



为了避免人员伤害和设备损坏，如果泵被放置在桌子上，请确保其稳定性。如果泵入口未连接至系统或完全堵塞，则切勿运行涡轮泵。

注意!

TwisTorr 305 FS 板载控制器属于 EN 61010-1 标准的第二安装（或过电压）类别。因此，请将设备连接到符合上述类别的电源线。TwisTorr 305 FS 板载控制器具有输入/输出和串行通信连接器，必须将其连接到外部电路，以使电子板的带电部件无法被接触到。根据 EN 61010-1 标准，即使发生单个故障，也要确保连接到 TwisTorr 305 FS 板载控制器的设备充分绝缘。

有关可选配件的安装，请参阅“技术信息”。

使用

本段详细介绍了主要操作程序。请参考涡轮泵说明手册，在操作涡轮控制器之前，进行所有真空歧管和电气的连接。

TwisTorr 305 FS 板载控制器的启动和使用

如要启动TwisTorr 305 FS板载控制器，只需提供电源即可。控制器会自动识别联锁和启动信号的存在并启动泵。

泵第一次开机时，控制器会在“慢启动”模式下自动启动泵。在关闭泵之前，请等待慢启动完成（直到LED发出稳定的绿灯光来指示泵已达到最大转速）为止。

通过“慢启动”模式启动泵，轴承内的润滑脂将以正确的方式进行重新分配（在长时间不使用泵之后必须这样做）。

如上所述，第一次启动泵时，控制器会自动执行慢启动。随后，如果泵在60天的时间内保持不活动状态，则下次启动泵时，用户必须以“慢启动”模式启动泵。

如要在激活“慢启动”的情况下启动泵，必须在软件中或通过远程I/O连接器启用慢启动模式。

有关“慢启动”模式的激活过程，请参见“技术信息”章节中的“信号说明”一节。TwisTorr 305 FS板载控制器上绿色LED灯的闪烁速度表示系统运行状况：

- 无闪烁：泵正常旋转；
- 缓慢闪烁（大约400毫秒）：系统处于突起加减速、停止或“等待互锁”的状态；
- 快速闪烁（大约200毫秒）：错误情况。

TwisTorr 305 FS 板载控制器的关闭

如要关闭TwisTorr 305 FS板载控制器，只需断开电源即可。集成控制器会立即停止泵。

紧急停止

如要在紧急情况下立即停止TwisTorr 305 FS板载控制器，请拔下电源线。

维护

TwisTorr 305 FS板载控制器不需要任何维护。在系统上执行的任何工作都必须由授权人员执行。在系统上进行任何操作之前，必须将其与电源断开连接。如果发生故障，可以使用安捷伦维修服务。可通过安捷伦预先更换服务来替换控制器。

警告!



在控制器上进行任何操作之前，请先断开其电源。

注意

在将控制器退还给制造商进行维修或更换翻新设备之前，必须填写本说明手册所附的“退货请求”表格并将其发送给当地的销售办事处。

在寄出之前，必须将表格的副本放入系统包装中。

如果要废弃控制器，则必须按照特定的国家标准进行处置。

处置方式

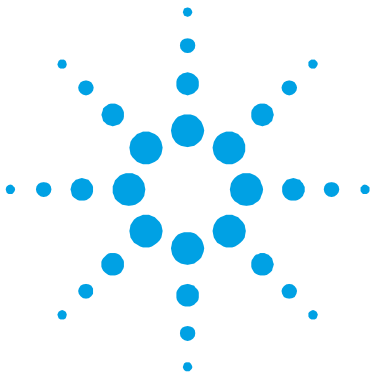
标签中的"WEEE"徽标的含义

以下符号根据 EC WEEE（废弃电气和电子设备）指令进行应用。该符号（**仅在欧洲共同体国家有效**）表示其适用的产品不得与普通家庭或工业废物一起处理，而必须送往差别化废物收集系统。因此，请最终用户在检查合同销售条款和条件后，联系设备的供应商（无论是母公司还是零售商）启动收集和处置流程。



有关详细信息，请参阅：

<http://www.agilent.com/environment/product/index.shtml>



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取り扱い説明書原版の翻訳



概要情報

この装置は技術者による使用を対象としています。使用者は、この取扱説明書とAgilentにより提供される他の追加情報もあわせて、装置を操作する前に全てお読みください。Agilentは、部分的であってもこれらの取扱説明に従わない場合や、訓練されていない人による不適切な使用、装置への認められていない修理、または特定の国家規格の規定に相いれないいかなる行動によって生じたいかなる事態にも責任を負いません。

TwisTorr 305 FS オンボードコントローラーは、TwisTorr 305 ポンプとの使用に適しています。追加のファンに電力を供給したり、通気弁を制御したり、シリアルライン(RS232 または RS485)に接続されたホストコンピュータによってTwisTorr 305 FS オンボードをリモート操作したりできるようにする補助コネクタと共に装備されています。

下記の章は、装置の使用時に操作者の安全を保証するために必要なすべての情報が含まれています。詳細情報は、「技術情報」の追記に記載されています。

この説明書では下記の表記が採用されています：

注意!

「注意」のメッセージは、もし遵守されていない場合に装置に損害が生じうる操作手順の前に表示されます。

警告!



「警告」のメッセージは、もし正しく操作されない場合に、重大な損傷につながる可能性がある特定の手順や実行に対する操作者の注意の必要性を示しています。

注

「注」は、テキストからの重要な情報を含みます。

保管

コントローラーを輸送、保管する時は、下記の環境仕様を超過してはいけません：

- 温度範囲： -20 °C から +70 °C まで
- 相対湿度範囲： 0 から 95 %まで（結露なし）

取り付けの準備

TwisTorr 305 FS オンボード は特殊な保護梱包で提供されます；もしこの梱包に移送の間に生じた可能性のある破損が見受けられる場合、現地販売事務所に問い合わせてください。不適切な方法で梱包材料を破棄しないでください。梱包材料は100%リサイクルが可能で、環境保護に関するEEC指令85/399に従っています。

取り付け

大気物質（雨、雪、氷）、ちり・ホコリ、侵略的ガスなどが露出する環境、または爆発の可能性がある環境、または燃えやすい環境下では、取り付けまたは使用しないでください。操作中、下記の環境状況は遵守されなければいけません：

- 温度： +5 °C から+45 °Cまで；
- 相対湿度： 0 – 95 %（結露なし）

警告!



使用者の安全のために、ターボコントローラー（主電源電圧のバージョンで： **X3507-64131**）は 3本のワイヤー電源コード（注文可能な部品を参照）とプラグ（国際的に認可されたもの）で電源が供給されなければいけません。電気ショックを避け、また CE 要件を満たすために、常にこの電源コードを使用し、適切な接地接続の電源ソケットにプラグを挿入してください。コントローラー内で発達した高電圧は、重大な損傷や死亡を引き起こす可能性があります。ユニットを取り付ける前または修理する前には常に、ソケットから接続を断ってください

警告!



人体への損傷や装置への損害を避けるために、もしポンプがテーブルの上に置かれている場合には安定していることを確かめてください。もしポンプ吸入口がシステムに接続されていない、または抜け落ちている場合には、ターボポンプを決して操作しないでください。

注意!

TwisTorr 305 FS オンボードは、EN 61010-1 規格の第 2 取り付け（または過電圧）カテゴリに属します。したがって、上記のカテゴリに準拠する主電源ラインに機器を接続してください。TwisTorr 305 FS オンボードには、電子ボードの通電部分にアクセスできないように外部回路に接続する必要がある入力/出力およびシリアル通信コネクタがあります。EN 61010-1 規格による単一故障の場合でも、の TwisTorr 305 FS オンボードに接続されている機器の分離が十分であることを確認してください。

オプションのアクセサリ取り付けに関しては、「技術情報」を参照してください。

使用

この章は、主な操作手順について詳しく説明します。ターボコントローラーを操作する前に、ターボポンプの取扱説明書を参照して、すべての真空マニホールドと電気接続を行ってください。

TwisTorr 305 FS オンボードの使用とスイッチを ON にします

TwisTorr 305 FS オンボードのスイッチをオンにするには、主電源を供給するだけで十分です。コントローラーは自動的にインターロックと始動信号の存在を認識し、ポンプを始動します。

ポンプが電源オンになる初回時は、コントローラーによって「ソフトスタート」モードで自動的に起動されます。ソフトスタートが完了するまで（つまり、ポンプが最大回転速度に達したことを示すLEDが一定の緑色に光るまで）待ってから、ポンプの電源を切ります。

ポンプをソフトスタートモードで起動することにより、ベアリング内の潤滑油が正しい方法で再分配されます（ポンプが長期間非稼働である後に必要です）。

前述のように、ポンプが初回に起動した時に、コントローラーはソフトスタートを自動的に始動します。その後、もしポンプが60日間非稼働のままである場合は、次回起動するときに、ユーザーはソフトスタートモードで起動しなければなりません。

「ソフトスタート」をアクティブにしてポンプを始動するには、ソフトウェアで、またはリモートI/Oコネクタでソフトスタートモードを有効にする必要があります。

「ソフトスタート」モードのアクティブ化手順については、「技術情報」の章の「信号の説明」を参照してください。

TwisTorr 305 FS オンボードのフロントパネル上の緑色のLEDの点滅速度は、システムの動作状態を示します：

- 点滅なし：ポンプは正常に回転しています。
- 低速の点滅（約400 msの周期）：システムはランプ、停止、または「インターロック待機中」の状態です。
- 高速の点滅（約200 msの周期）：エラーの状態です。

TwisTorr 305 FS オンボードのスイッチを OFF にします

TwisTorr 305 FS オンボードのスイッチをオフにするには、主電源を切るだけで十分です。統合されたコントローラーはポンプを即座に停止します。

緊急停止

緊急事態にTwisTorr 305 FS オンボードを即座に停止するには、電源コードを取り外してください。

メンテナンス

TwisTorr 305 FS オンボードはいかなるメンテナンスも必要としません。システムになされるいかなる作業も、認可された担当者により実行されなければなりません。

システムにいかなる操作を実行する前には、必ず電源を切断してください。もし故障が発生した場合は、Agilentの修理サービスを利用することもできます。交換用コントローラーは、Agilentを通じてアドバンスエクスチェンジをもとに入手できます。

警告!



コントローラー上にいかなる作業を実行する前には、電源から接続を断ってください。

注

コントローラーを修理のためにメーカーに返却する前、または再生ユニットと交換する前に、この取扱説明書に添付されている「返品依頼」シートに記入して、現地販売事務所へ送付する必要があります。郵送する前に、シートのコピーをシステムのパッケージに挿入してください。

もしポンプを廃棄する場合には、特定の国の規定に基づき廃棄されなければなりません。

廃棄

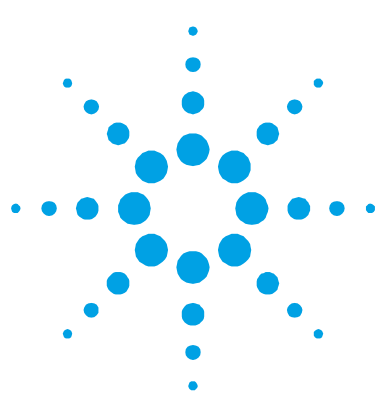
ラベル上に表示された「WEEE」ロゴの意味

下記の記号はEC WEEE指令（電気電子廃棄物 指令）に従い適用されたものです。この記号（欧州諸共同体でのみ有効）は、製品は通常の家
庭ゴミまたは産業廃棄物と一緒に廃棄されてはいけないこと、また分
化された廃棄物収集システムへ送られなければならないことを示しま
す。従って、エンドユーザーは、販売の契約条件を確認した後に、収
集と廃棄工程を始めるために、親会社かもしくは小売業者であらう
と、機器の供給者へ問い合わせることが推奨されます。



さらなる情報は下記をご確認ください:

<http://www.agilent.com/environment/product/index.shtml>



7

Instructions for Use

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Original Instructions



General Information

This equipment is destined for professional use. Before operating the equipment, the user should read this instruction manual and any other additional information supplied by Agilent. Agilent declines any responsibility for non-compliance, even partial, with these instructions, improper use by untrained persons, unauthorized repairs of the equipment or use not complying with specific national standards.

The TwisTorr 305 FS onboard controller is suitable for use with TwisTorr 305 pumps. It is equipped with auxiliary connectors that make it possible to power an additional fan, to control the vent valve and for remote operation of the TwisTorr 305 FS onboard by a host computer connected on a serial line (RS232 or RS485). The following paragraphs contain all the information necessary to guarantee operator safety when using the equipment. For detailed information, refer to the "Technical Information" appendix.

The following conventions are adopted in this manual:

CAUTION!

Caution messages are displayed before procedures which, if not followed, may cause damage to the equipment.

WARNING!



Warning messages are intended to draw the operator's attention to a particular procedure or practice which, if not followed correctly, may cause serious injury.

NOTE

The notes contain important information taken from the text.

Storage

When transporting and storing the pumps, the following environmental conditions should not be exceeded:

- temperature: from -20 °C to +70 °C
- relative humidity: 0 – 95 % (non-condensing)

Preparation for Installation

The TwisTorr 305 FS onboard is shipped in a special protective packaging; in the case of damage during transport, contact your local sales centre. Do not dispose of the packaging in the environment. The material is totally recyclable in accordance with Directive 85/399/EEC on environmental protection.

Installation

Do not install or use the controller in an environment exposed to atmospheric agents (rain, snow, ice), dust, aggressive gases, or in explosive environments or those with a high fire risk. During operation, the following environmental conditions must be complied with:

- temperature: from + 5 °C to +45 °C
- relative humidity: 0 – 90 % (non-condensing)

WARNING!



For user safety, the Turbo controller (in grid voltage version, X3507-64131) must be powered by a 3-wire power cord (see orderable parts table) and plug (internationally approved). Always use this power cord and insert the plug in a socket with a suitable ground connection to avoid electrical shock and to comply with EC requirements. The high voltage developed inside the controller can cause severe injury or death. Before installing or servicing the unit, always disconnect it from the socket.

WARNING!



To avoid injury to personnel and damage to the equipment, if the pump is lying on a table, make sure it is steady. Never operate the Turbo pump if the pump inlet is not connected to the system or blanked off.

CAUTION!

The TwisTorr 305 FS onboard belongs to the second installation (or overvoltage) category of the EN 61010-1 standard. Therefore, connect the device to a mains line that complies with the above category. The TwisTorr 305 FS onboard has Input/Output and serial communication connectors that must be connected to external circuits in such a way that no live parts of electronic board are accessible. Make sure that the insulation of the device connected to the TwisTorr 305 FS onboard is adequate even in the case of a single fault as per the EN 61010-1 standard.

For installation of optional accessories, see "Technical Information".

Use

This paragraph details the main operating procedures. Prior to operating the Turbo controller, make all vacuum manifold and electrical connections, referring to the Turbo pump instruction manual.

Switching on and Use of the TwisTorr 305 FS onboard

To switch on the TwisTorr 305 FS onboard, it is sufficient to provide the mains supply. The controller automatically recognizes the presence of the interlock and start signals and starts the pump.

The first time the pump is powered on, it is started automatically in “Soft Start” mode by the controller. Wait until the Soft Start completes (that is, until the LED emits a fixed green light indicating that the pump has reached its maximum rotation speed), before switching off the pump.

By starting the pump in Soft Start mode, the grease inside the bearings is redistributed in the correct manner (necessary after the pump has been inactive for a long period).

As stated, the first time the pump is started, the controller executes Soft Start automatically. Subsequently, if the pump remains inactive for a period of time of 60 days, the next time it is started the user must start it in Soft Start mode.

To start the pump with “Soft Start” activated, Soft Start mode must be enabled in the software or through the Remote I/O connector.

For the “Soft Start” mode activation procedure, see the “Signal Description” paragraph in the “Technical Information” chapter. The flashing rate of the green LED on the TwisTorr 305 FS onboard front panel indicates system operating conditions:

- with no flashing: the pump is rotating normally;
- slow flashing (period of about 400 ms): the system is in ramp, or in Stop, or in “Waiting for interlock” status;
- fast flashing (period of about 200 ms): error condition.

Switching off the TwisTorr 305 FS onboard

To switch off the TwisTorr 305 FS onboard, it is sufficient to remove the mains supply. The integrated controller immediately stops the pump.

Emergency Stop

To stop the TwisTorr 305 FS onboard immediately in an emergency condition, remove the power cord.

Maintenance

The TwisTorr 305 FS onboard does not require any maintenance. Any work performed on the system must be carried out by authorized personnel. Before carrying out any operation on the system, it must be disconnected from the power supply. If a fault occurs, it is possible to use the Agilent repair service. Replacement controllers are available on an advance exchange basis through Agilent.

WARNING!

Before carrying out any work on the controller, disconnect it from the power supply.

NOTE

Before returning the controller to the manufacturer for repairs, or replacement with a reconditioned unit, the "Request for return" sheet attached to this instruction manual must be filled in and sent to the local sales office. A copy of the sheet must be inserted in the system packaging before shipping.

If a controller is to be scrapped, it must be disposed of in accordance with the specific national standards.

Disposal

Meaning of the "WEEE" logo found in labels.

The following symbol is applied in accordance with the EC WEEE (Waste Electrical and Electronic Equipment) Directive.

This symbol (**valid only in countries of the European Community**) indicates that the product it applies to must NOT be disposed of together with ordinary domestic or industrial waste but must be sent to a differentiated waste collection system. The end user is therefore invited to contact the supplier of the device, whether the Parent Company or a retailer, to initiate the collection and disposal process after checking the contractual terms and conditions of sale.



For more information refer to:

<http://www.agilent.com/environment/product/index.shtml>



8 Technical Information

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8 Technical Information

Disposal

Controller Description

Two different versions of the controller are available:

X3507-64130	TwisTorr 305 FS onboard controller 24V
X3507-64131	TwisTorr 305 FS onboard controller 110/230V

The controller is a solid-state frequency converter with the following features:

- Drives the pumps of the TwisTorr 305 FS family.
- Powers the pump cooling fan.
- Drives the vent valve.
- Remote I/O compatible with the previous version.
- Serial communication compatible with the previous RS 232 and 485 versions; due to some new features which are present in the software, some minor changes to the serial protocol have been made.

NOTE

With reference to the previous 304 and 301 pump families, the serial “window protocol” differences implemented for the 305 family are the following:

- Added WIN001, WIN143, WIN144, WIN167, WIN157
- Removed WIN106, WIN107

Refer to the “Technical Information – window meanings” chapter to get the relevant details.

The dedicated controller is a solid-state frequency converter which is driven by a single chip microcomputer and consists of two PCBs which include power supply and 3-phase output, analog and input/output section, microprocessor and digital section. The power supply, together with the 3-phase output, converts the single phase AC mains supply or 24Vdc supply into a 3-phase, low voltage, medium frequency output which is required to power the pump. The controller can be operated by a remote host computer via the serial connection. A Windows-based software is available (optional).

Technical Specification

Table 1 Technical Specification

Pwr supply (mains):	
Input voltage:	100 - 240 Vac
Input freq.:	50 - 60 Hz
Max input pwr:	300 VA
Stand-by pwr:	10 W
Max oper. power:	150 W with water or air cooling
Protection fuse	4 A
Pwr supply (24 Vdc):	
Input voltage:	24 Vdc
Max input pwr:	200 W
Stand-by pwr:	10 W
Max oper. power:	150 W with water or air cooling
Protection fuse	8 A
Compliance with:	UNI EN 292-1 UNI EN 292-2 EN-CENELEC 55011 IEC 1000-4-2 (ex 801-2) IEC 1000-4-3 (ex 801-3) IEC 1000-4-4 (ex 801-4) EN 61010-1 (IEC 1010-1) EN 1012-2
Power cable	With European or NEMA plug 3 meters long (optional)
Serial communication (TwisTorr 305 kit)	RS232 cable with a 9-pin D type male connector and a 9-pin D type female connector, and TwisTorr software (optional)
Storage temperature	-20 ° C to +70 ° C
Input	75 Vac, three phase, 1010 Hz
Internal use only	Max. altitude 2000 m
Installation category	II

Pollution degree	2
Operating ambient temperature	5 ° C to 40 ° C
Weight	1 kg (2,2 lbs)

NOTE

When the TwisTorr 305 FS onboard has been stored at a temperature less than 5°C, wait until the system has reached the above mentioned temperature.

Controller Outline

The outline dimensions for the TwisTorr 305 FS onboard controllers are shown in the following figure.

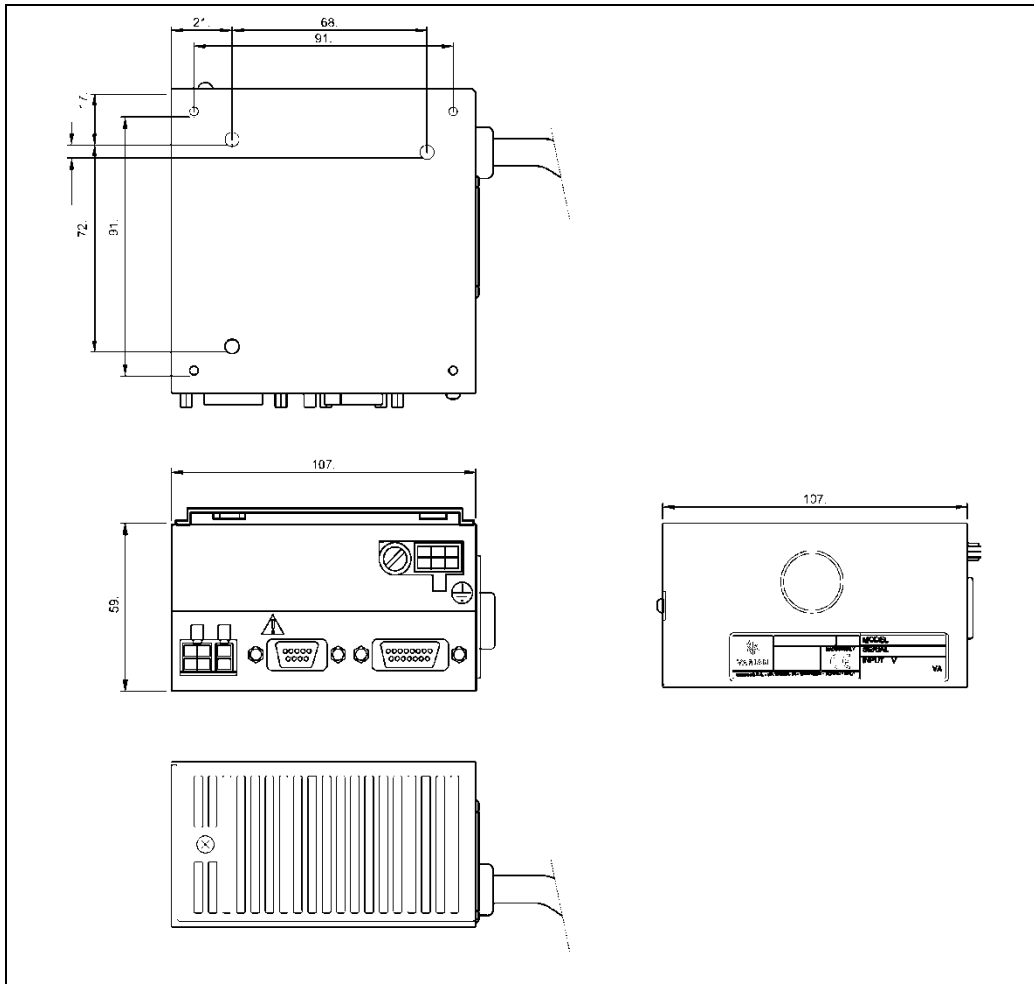


Figure 1 Outline 24 V – X3507-64130

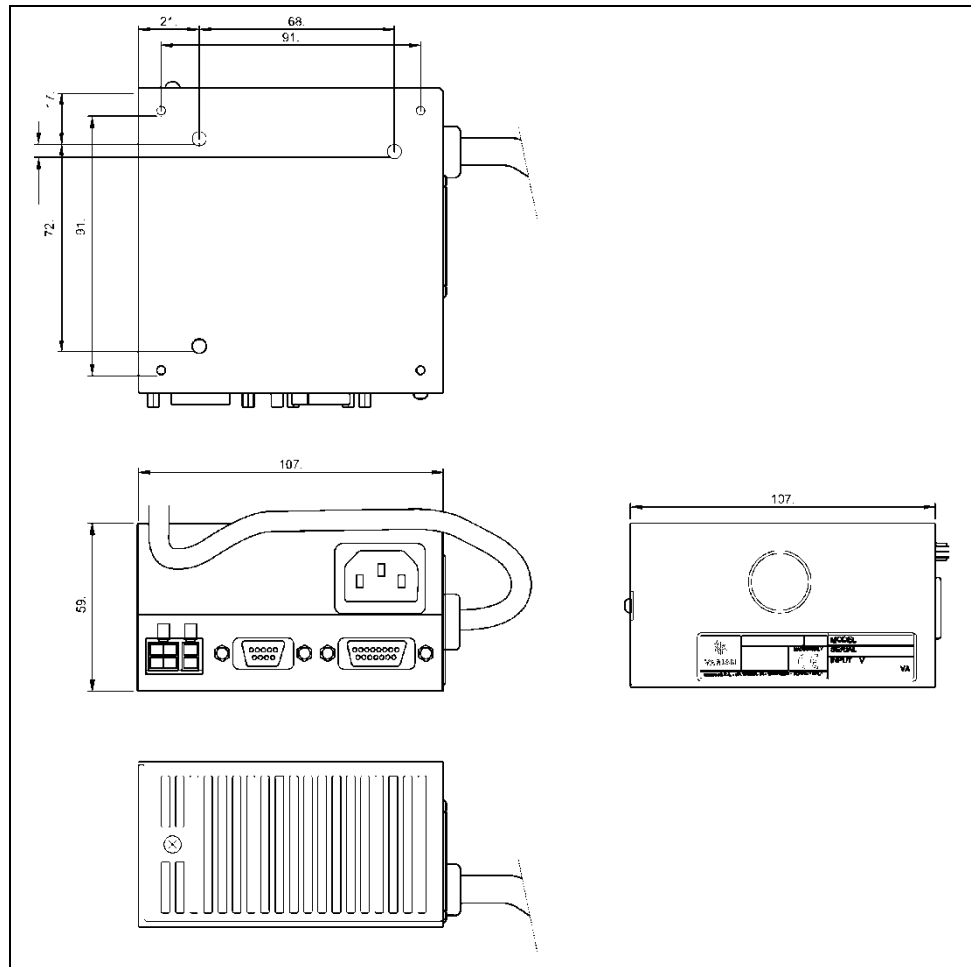


Figure 2 Outline X3507-64131

Interconnections

The following figure shows the TwisTorr 305 FS onboard interconnections.

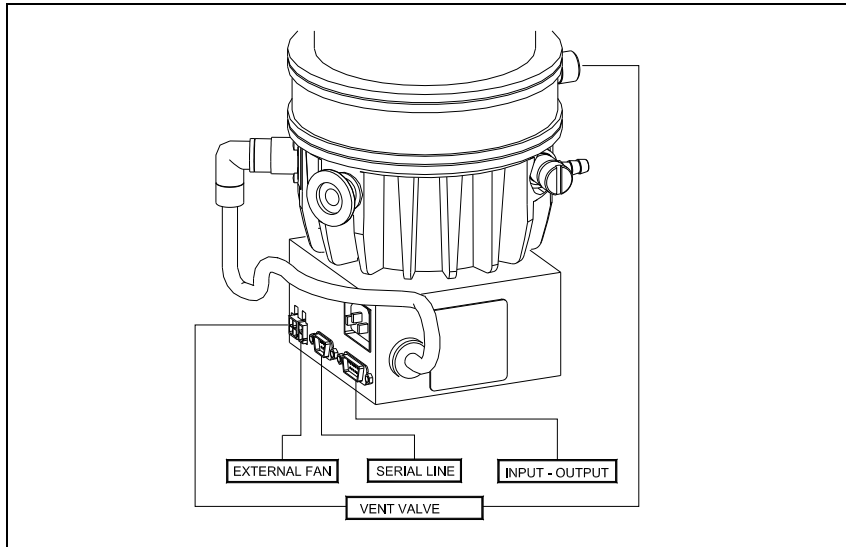


Figure 3 TwisTorr 305 FS onboard interconnections

P3 - Vent

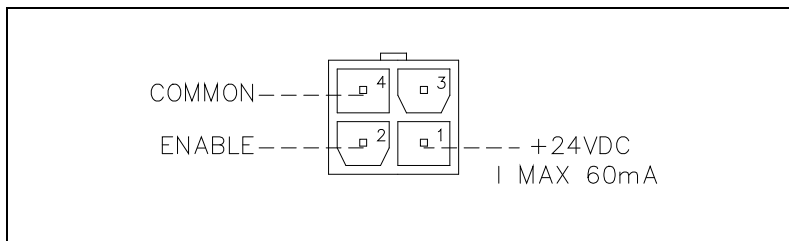


Figure 4 P3-Vent

This is a dedicated 24 Vdc connector to control the optional vent valve.

P5 – External Fan

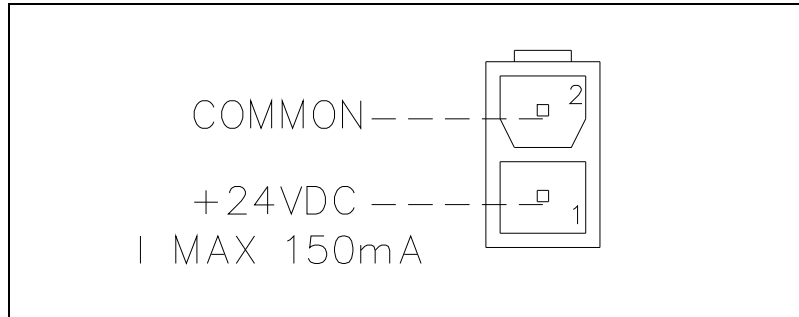


Figure 5 External fan

This is a dedicated 24 Vdc connector to supply the optional external fan.

J5 – In-Out

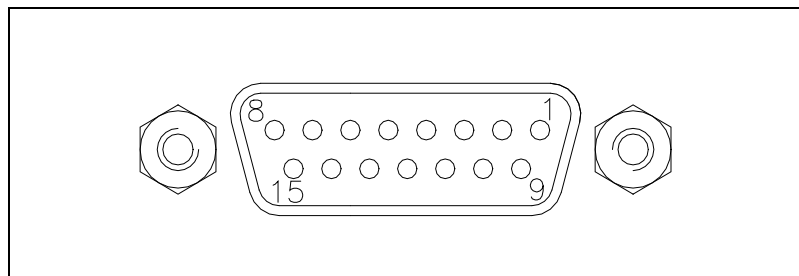


Figure 6 In-Out

This connector carries all the input and output signals to remote control the TwisTorr 305 FS onboard.

It is a 15-pins D type connector; the available signals are detailed in the table, the following paragraphs describe the signal characteristics and use.

Table 2 Signal characteristics and use

PIN N.	SIGNAL NAME	IN-/OUTPUT
1	Start/Stop (+)	In
2	Start/Stop (-)	In
3	Interlock (+)	In
4	Interlock (-)	In
5	Speed setting (+)	In
6	Speed setting (-)	In
7	Soft start (+)	In
8	Soft start (-)	In
9	+ 24 vdc	Out
10	Spare	
11	Programmable set point (+)	Out
12	Contact: normally open	
13	Fault output	Out
14	Programmable analog signal (+)	Out
15	• Ground • Programmable analog signal (-)	Out

Signal Description

Start/Stop: input signal to start or stop the pump. With the supplied cover connector the START/STOP (+) signal is connected to the +24 Vdc pin and the START/STOP (-) signal to the GROUND pin: in this condition the pump automatically starts as soon as the controller recognises the input supply ("Plug & Pump").

Interlock: input signal to control the pump rotation. With the supplied cover connector the interlock (+) signal is connected to the +24 Vdc pin and the interlock (-) signal to the GROUND pin.

Soft start: this input is used to provide a "soft start" to the pump; in this condition the ramp-up time could be up to 45 min.

Speed setting: PWM input signal to set the pump speed. The PWM signal characteristics must be the following:

- frequency: 100 Hz +/-20%
- amplitude: from 5 to 24 V
- duty cycle range: from 25% to 75% (t_{off}/T) corresponding to a rotational frequency from 800 Hz to 1010 Hz linearly. With duty cycle <25% rotational frequency = 800 Hz, duty cycle >75% rotational frequency = 1010 Hz

NOTE

The duty cycle percentage is referred to the low level portion of the PWM signal.

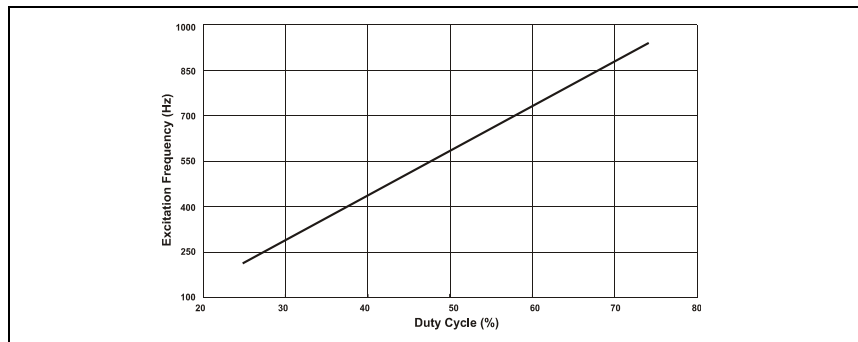


Figure 7

Programmable analog signal: this output signal is a voltage (from 0 to 10 Vdc) proportional to a reference quantity (frequency or power) set by the user. The default setting is the frequency (see the following example diagram).

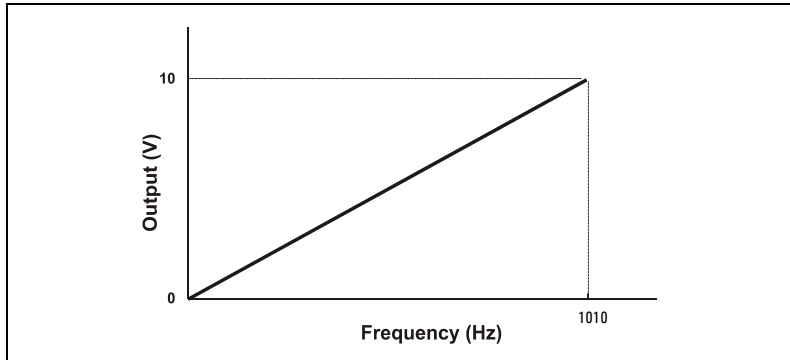


Figure 8

Fault: this open collector output signal is ON when a system fault condition is detected.

Programmable set point: this open collector output signal is enabled when the reference quantity chosen (frequency, current or time) is higher than the set threshold. The signal can be "high level active" (that is the output is normally at 0 Vdc and becomes 24 Vdc when activated), or "low level active" (that is the output is normally at 24 Vdc and becomes 0 Vdc when activated). Moreover, if the reference quantity is the frequency or the current drawn, it is possible to set the hysteresis (in % of the threshold value) to avoid bouncing.

For example:

- reference quantity: frequency
- threshold: 500 Hz
- hysteresis: 1%
- activation type: "high level"

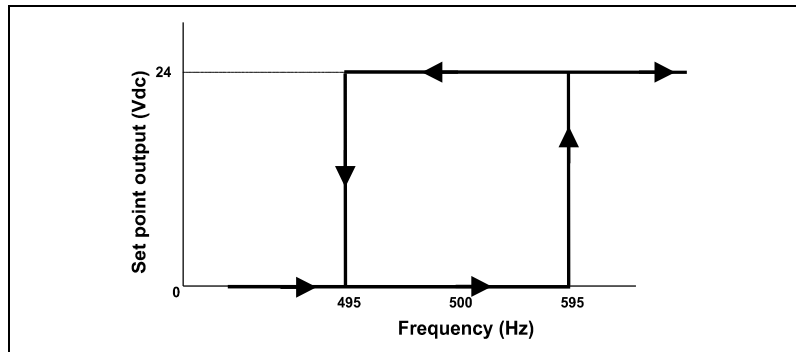


Figure 9 Set point output

The set point output stays at 0 Vdc until the frequency becomes higher than 505 Hz (that is 500 Hz + 1% of 500 Hz), then the output goes at 24 Vdc and stays at 24 Vdc until the frequency becomes lower than 495 Hz (that is 500 Hz – 1% of 500 Hz).

It is possible to delay the set point checking for a programmable delay time.

The PROGRAMMABLE SET POINT signal has the following default settings:

- reference quantity: frequency
- threshold: 909 Hz
- hysteresis: 2 %
- activation type: high level
- delay time: 0 second

NOTE

The communication S/W between the controller and the pump (A-Plus) allows the operator to set all the programmable feature.

When no external input-output device is available this connector must be closed with the supplied mating connector that short-circuits the START and INTERLOCK inputs with the GROUND input (see the following figure).

8 Technical Information

How to Connect the Open Collector Inputs of the Controller

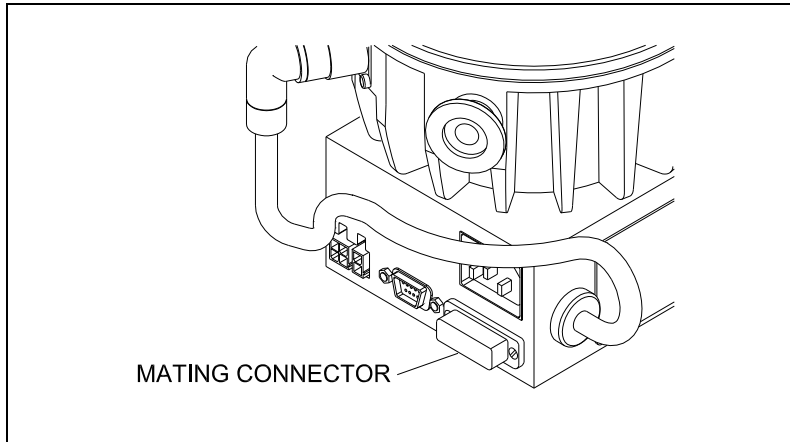


Figure 10

How to Connect the Open Collector Inputs of the Controller

Here below there are the typical connections of the open collector input of the integrated controller to an external system. Two cases are considered:

1. the customer supplies the 24 Vdc
2. the customer does not supply the 24 Vdc

Please note that on the connector a 24 Vdc, 60 mA voltage, a GROUND signal and the open collector pin are available.

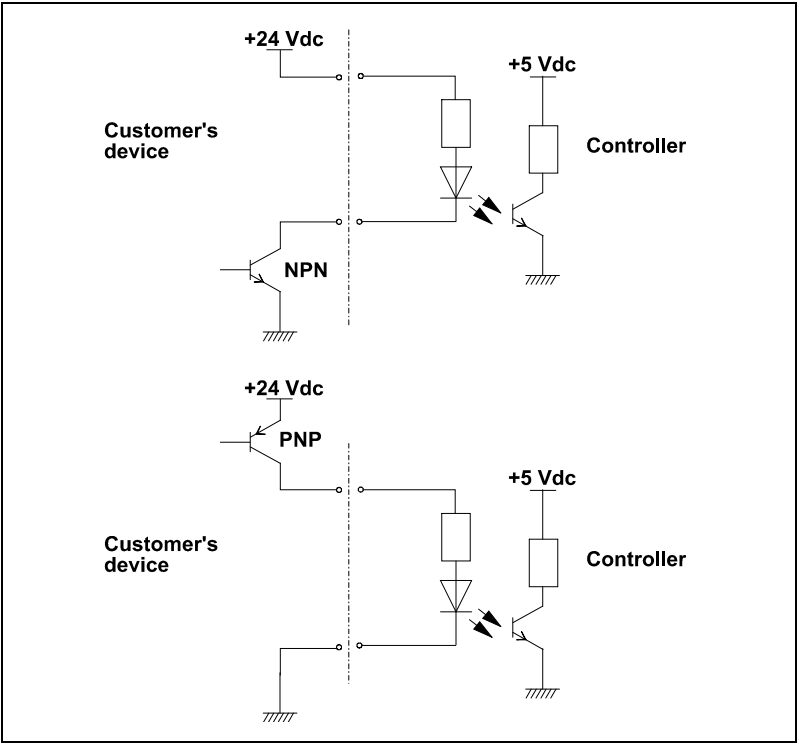


Figure 11 Case 1

8 Technical Information

How to Connect the Open Collector Inputs of the Controller

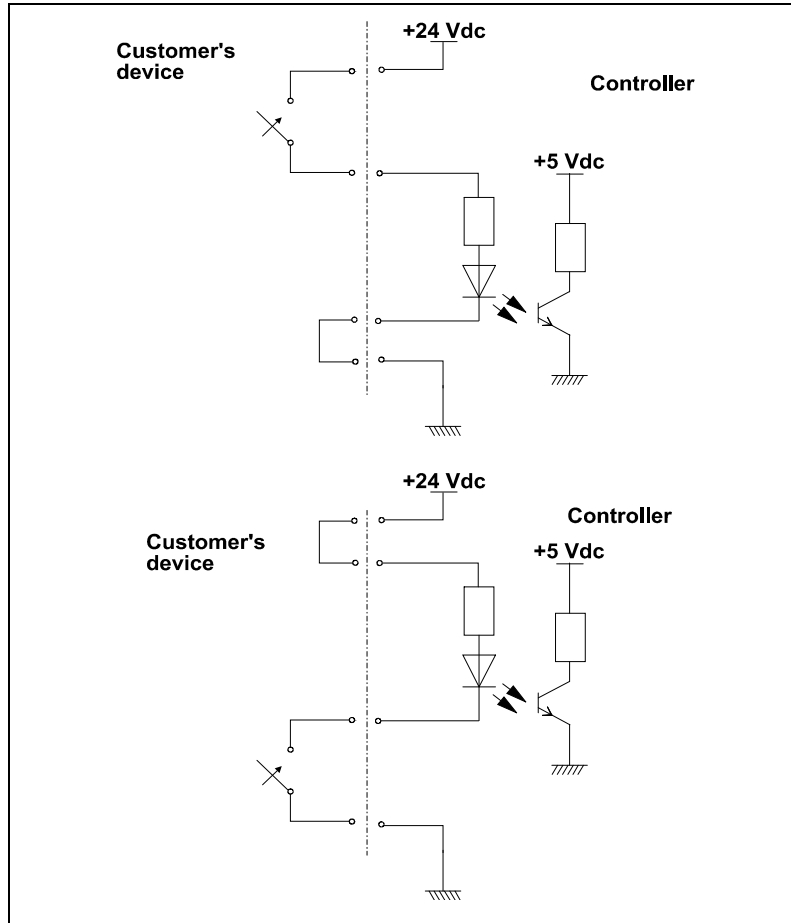


Figure 12 Case 2 with relay utilisation

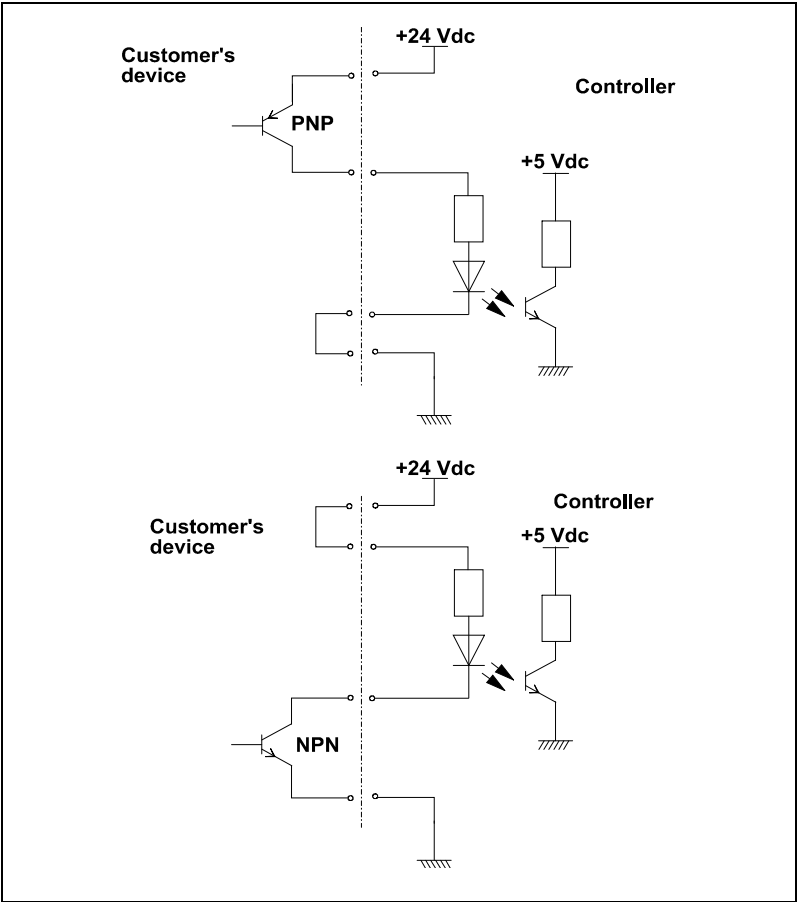


Figure 13 Case 2 with transistor utilisation

How to Connect the Outputs of the Controller

The following figure shows a typical logic output connection (relay coil) but any other device may be connected e.g. a LED, a computer, etc., and the related simplified circuit of the controller. The figure example refers to the programmable set point signal on pins 11 and 9.

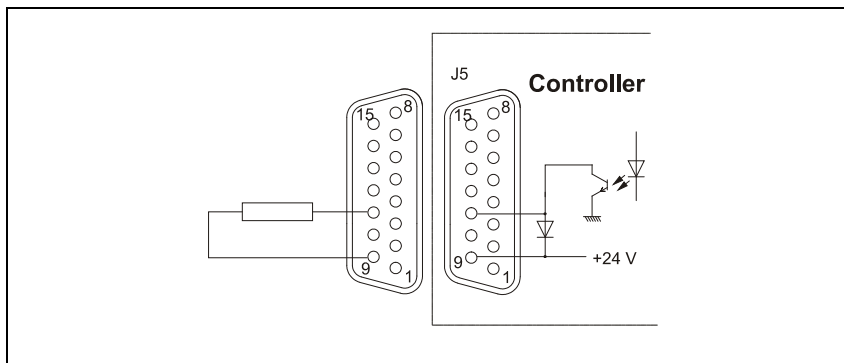


Figure 14

J6 – Serial

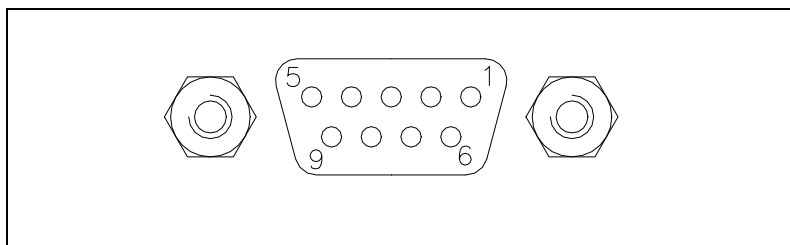


Figure 15 9 Pin connector

This is a 9 pin D-type serial input/output connector to control via an RS 232 or RS 485 connection the TwisTorr 305 FS onboard.

Table 3 Signal characteristics and use

PIN N.	SIGNAL NAME
1	+5 V (OUT)
2	TX (RS232)
3	RX (RS232)
4	A+ (RS422 Optional)
5	GND
6	A + (RS485)
7	B - (RS422 Optional)
8	B - (RS485)
9	RESERVED

NOTE

Note that the vent valve can also be controlled by means of the serial connection.

A serial communication kit with a serial cable and the T-Plus software is available (optional).

Input power connection on the model X3507-64130

The following figure shows the input power connection relevant to the model n. X3507-64130.

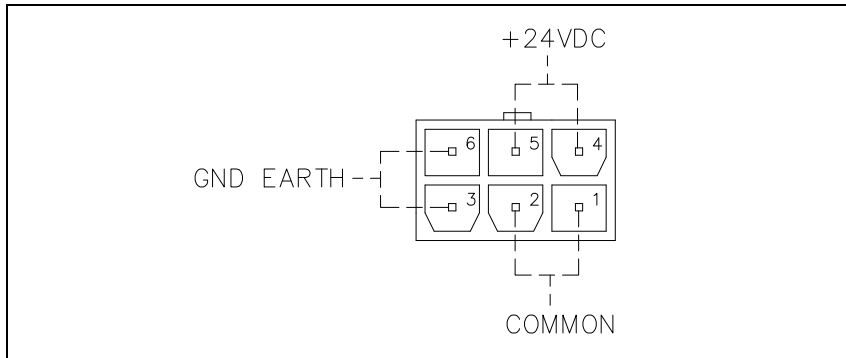


Figure 16

RS 232/RS 485 communication description

Both the RS 232 and the RS 485 interfaces are available on the connector J6.

The communication protocol is the same (see the structure below), but only the RS 485 manages the address field. Therefore to enable the RS 485 is necessary to select the type of communication as well as the device address by means of the TwisTorr software.

Communication Format

- 8 data bit
- no parity
- 1 stop bit
- baud rate: 600/1200/2400/4800/9600 pro-grammable

Communication Protocol

The communication protocol is a MASTER/SLAVE type where:

- Host = MASTER
- Controller = SLAVE

The communication is performed in the following way:

1. the host (MASTER) send a MESSAGE + CRC to the controller (SLAVE);
2. the controller answer with an ANSWER + CRC to the host.

The MESSAGE is a string with the following format:

<STX>+<ADDR>+<WIN>+<COM>+<DATA>+<ETX>+<CRC>

where:

NOTE

When a data is indicated between two quotes ('...') it means that the indicated data is the corresponding ASCII character.

- <STX> (Start of transmission) = 0x02
- <ADDR> (Unit address) = 0x80 (for RS 232)
<ADDR> (Unit address) = 0x80 + device number (0 to 31) (for RS 485)
- <WIN> (Window) = a string of 3 numeric character indicating the window number (from '000' to '999'); for the meaning of each window see the relevant paragraph.
- <COM> (Command) = 0x30 to read the window, 0x31 to write into the window
- <DATA> = an alphanumeric ASCII string with the data to be written into the window. In case of a reading command this field is not present. The field length is variable according to the data type as per the following table:

Table 4

DATA TYPE	FIELD LENGTH	VALID CHARACTERS
Logic (L)	1	'0' = off '1' = on
Numeric (N)	6	'-', '.', '0' ... '9' right justified with '0'
Alphanumeric (A)	10	from blank to '_' (ASCII)

- <ETX> (End of transmission) = 0x03
- <CRC> = XOR of all characters subsequent to <STX> and including the <ETX> terminator. The value is hexadecimal coded and indicated by two ASCII character.

The addressed SLAVE will respond with an ANSWER whose structure depends from the MESSAGE type.

When the MESSAGE is a reading command, the SLAVE will respond transmitting a string with the same structure of the MESSAGE.

NOTE

Using the RS 485 interface, the message structure remains identical to the one used for the RS 232 interface, the only difference being that the value assigned to the ADDRESS <ADDR>

The controller can answers with the following response types:

Table 5 Response types

Type	Length	Value	Description
Logic	1 byte	-	After a read instruction of a logic window
Numeric	6 bytes	-	After a read instruction of a numeric window
Alphanumeric	10 bytes	-	After a read instruction of an alphanumeric window
ACK	1 byte	(0x6)	The command execution has been successfully completed

Type	Length	Value	Description
NACK	1 byte	(0x15)	The command execution has been failed
Unknown Window	1 byte	(0x32)	The specified window in the command is not a valid window
Data Type Error	1 byte	(0x33)	The data type specified in the command (Logic, Numeric or Alphanumeric) is not accorded with the specified Window
Out of Range	1 byte	0x34)	The value expressed during a write command is out of the range value of the specified Window
Win Disabled	1 byte	(0x35)	The specified window is Read Only or temporarily disabled (for ex-ample you can't write the Soft Start when the Pump is running)

8 Technical Information

RS 232/RS 485 communication description

Examples:

Command: START

Source: PC

Destination: Controller

02	80	30	30	30	31	31	03	42	33
STX	ADDR	WINDOW			WR	ON	ETX	CRC	

Source: Controller

Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Command: STOP

Source: PC

Destination: Controller

02	80	30	30	30	31	30	03	42	32
STX	ADDR	WINDOW			WR	OFF	ETX	CRC	

Source: Controller

Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Command: SOFT-START (ON)

Source: PC
 Destination: Controller

02	80	31	30	30	31	31	03	42	32
STX	ADDR	WINDOW			WR	ON	ETX	CRC	

Source: Controller
 Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Command: SOFT-START (OFF)

Source: PC
 Destination: Controller

02	80	31	30	30	31	30	03	42	33
STX	ADDR	WINDOW			WR	OFF	ETX	CRC	

Source: Controller
 Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Window-Meanings

Table 6 Window-Meanings

N.	Read/Write	Datatype	Description	Admitted Values
000	R/W	L	Start/Stop (in remote mode the window is a read only)	Start = 1; Stop = 0 (default = 0)
001	R/W	L	Low Speed activation command	Active = 1; Non Active = 0 (default = 0)
008	R/W	L	Remote or Serial configuration	Remote = 1; Serial = 0 (default = 1)
100	R/W	L	Soft Start activation command (write only in Stop condition)	Active = 1; Non Active = 0 (default = 0)
101	R/W	N	Set Point type	0 = Frequency; 1 = Current; 2 = Time (default = 0)
102	R/W	N	Set Point threshold (expressed in Hz, mA or s)	(default = 909)
103	R/W	N	Set Point delay: time between the pump start and the set point check (seconds)	0 to 65535 (default = 0)
104	R/W	L	Set Point signal activation type: the signal can be "high level active" or "low level active"	0 = high level active; 1 = low level active (default = 0)
105	R/W	N	Set point hysteresis (in % of threshold)	Set point hysteresis, in % of threshold (default = 2)
108	R/W	N	Baud Rate	0 = 600 bit/s; 1 = 1200 bit/s; 2 = 2400 bit/s; 3 = 4800 bit/s; 4 = 9600 bit/s (default = 4)
109	W	L	Pump life/ cycle time/ cycle number reset	To reset the counters, write '1'
110	R/W	L	Interlock type. The interlock can be checked at start or continuously	Impulse = 0; Continuous = 1 (default = 1)

N.	Read/Write	Datatype	Description	Admitted Values
111	R/W	N	Analog output type: output voltage signal proportional to frequency or power	0 = frequency; 1 = power (default = 0)
120	R/W	N	Rotational frequency setting (Hz)	700 to 1010 (default = 1010)
121	R/W	N	Maximum rotational frequency in Hz (active only in Stop condition)	700 to 1010 (default = 1010)
122	R/W	L	Set vent valve on/off (on = closed)	On = 1; Off = 0 (default = 1)
125	R/W	N	Set the vent valve operation	Automatic = 0 (ref. to note 1) On command = 1 (ref. to note 2) (default = 0)
126	R/W	N	Vent valve opening delay (expressed in 0.2 sec)	0 to 65535 (corresponding to 0 to 13107 sec) (default = 15)
143	R/W	N	External fan configuration (always on or driven by serial command)	0 = Always On 1 = Serial (default = 0)
144	R/W	L	External Fan Setting (ON/OFF serial command)	On = 1; Off = 0 (default = 0) Not active if W143=0
157	R/W	N	Gas Type Selection	0 = Ar 1 = N2 2 = He (default = 0) Ref. to item 4 of the note.
167	R	L	Stop Speed Reading (it reads the pump speed while in stop mode)	On = 1; Off = 0 (default = 0)
200	R	N	Pump current in mA dc	
201	R	N	Pump voltage in Vdc	
202	R	N	Electrical power delivered to the pump in W (pump current x pump voltage duty cycle)	
203	R	N	Driving frequency in Hz	
204	R	N	Pump body temperature in °C	

8 Technical Information

RS 232/RS 485 communication description

N.	Read/Write	Datatype	Description	Admitted Values
205	R	N	Pump status	0 = Stop; 1 = Interlock; 2 = Ramp; 3 = Autotuning; 4 = Braking; 5 = Normal; 6 = Fail
206	R	N	Error code	Bit description: refer to the picture which is located on the bottom of this table
300	R	N	Cycle time in minutes (set to zero by a reset command)	0 to 999999
301	R	N	Cycle number (set to zero by a reset command)	0 to 9999
302	R	N	Pump life in hours (set to zero by a reset command)	0 to 999999
400	R	A	CRC EPROM (QE)	QE8XXXX (where "XXXX" are variable chars)
402	R	A	CRC Param. (PA)	PA8XXXX (where "XXXX" are variable chars)
404	R	A	CRC Parameter structure	"XXXX"
503	R/W	N	RS485 Address	0 to 31 (default = 0)
504	R/W	N	Serial type select	0 = RS 232; 1 = RS 485 (default = 0)
724	R/W	N	Run Up Time (in minutes)	0 to 30 (ref. to note 3) (default = 15)
725	R/W	L	Run Up Time activation command	On = 1; Off = 0 (default = 1)

NOTE

All the serial windows which are not mentioned in the relevant table are reserved for future use or reserved to Agilent Service.
These windows are not intended to be used by the end user.

NOTES

1. Automatic means that when the controller stops, the vent valve is opened with a delay defined by window n. 126; when the controller starts, the vent valve is immediately closed.
2. On command means that the vent valve is opened or closed by means of window n. 122.
3. Run Up Time is the maximum amount of time for which the pump is allowed to spin at a speed value less than the normal operating speed. The pump is not intended to spin for an indefinite time at a speed value lower than 700Hz. If the pump, due to the gas flow applied to it, spins at a speed value less than 700Hz, the Run Up timer starts to count; when the counter value exceeds the Run Up Time set value, the pump will stop and the error condition "Run Up Time" will be issued by the control unit.
4. Select the gas type which is appropriate for the customer process in order to achieve the full performance of the pump.

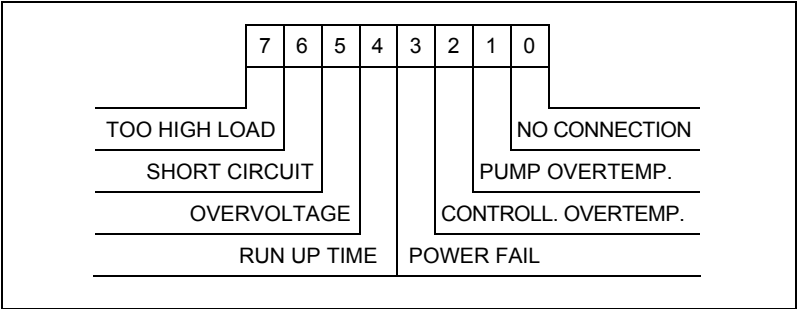


Figure 17 Window N. 206 Bit Description

Air Cooling Kit Installation

TwissTorr 305 FS onboard Controller

An air cooling kit (mod. X3500-68010) is available to improve the TV 301 cooling during heavy operational conditions (optional).

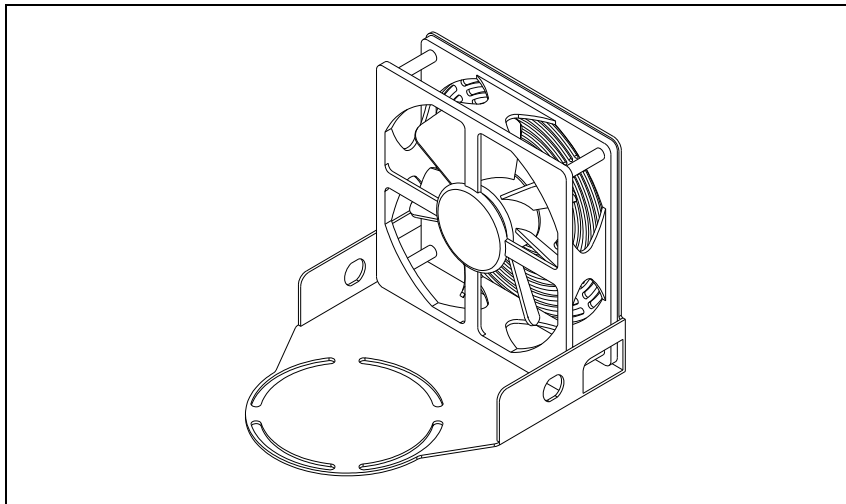


Figure 18

Fan specifications:

- air flow: 147 m³/h
- input voltage: 24 Vdc
- dimensions: 119 x 119 x 32 mm
- power: 2.6 W

The fan bracket is shaped so that it can be mounted close to the pump.

In case the pump is located far from the control unit and the control unit is connected to the pump via an extension cable, the cooling kit is already provided with a specific extension cable (which is included in its package) in order to connect it to the controller.

To fix the fan to the TwissTorr 305 FS onboard case execute the following procedure (see the following figure):

- 1 Fix the fan to the suitable bracket by means of the furnished screws;
- 2 Fix the bracket to the pump body between the pump and the controller;
- 3 Connect the fan supply to the P4 connector of the controller.

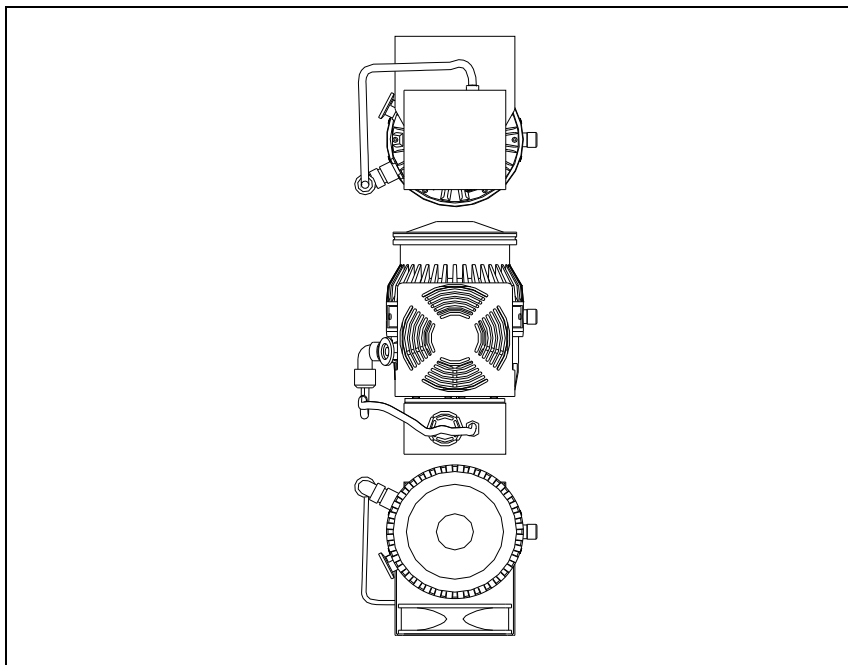


Figure 19

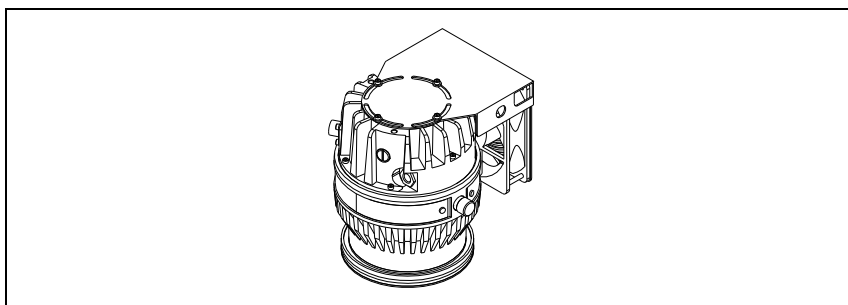


Figure 20

Use in the Presence of Magnetic Fields

Magnetic fields induce eddy currents in the rotor of a turbomolecular pump that tend to prevent rotation. This causes increased electrical power consumption by the motor, most of which is dissipated in the rotor. In general, therefore, an increase in pump current can be expected.

Accessories and Spare Parts

DESCRIPTION	PART NUMBER
Mains cable NEMA Plug, 3m long	9699958
Mains cable European Plug, 3m long	9699957
Serial cable	9699883
RS232 Serial to B/T Adapter	X3514-68003
Air cooling kit for TwisTorr 305 FS Controller	X3500-68010
Vent valve for TwisTorr 305 FS Controller	9699834

NOTE

The communication S/W between the controller and the pump (A-Plus) is available on the Agilent website:

<https://www.agilent.com/en/products/vacuum-technologies/vacuum-leak-detection-software/a-plus-software>

The connection is provided through serial communication or via RS232 SERIAL to B/T Adapter (X3514-68003).



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A handwritten signature in black ink, appearing to read "Giampaolo LEVI".

Giampaolo LEVI

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REFERENCE INFORMATION (model n°, serial n°, ordering information, time to failure after installation, etc.):		
DATE		
CORRECTIVE ACTION PLAN / ACTUATION (by AGILENT VPD)		LOG N°

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- o Products returned that have not been drained from oil will be disposed.
- o A Special Cleaning fee will apply to all exposed products.
- o If requesting a calibration service, units must be functionally capable of being calibrated.



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Failure description		Type of process (for which the equipment was used)	

Type of return	
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		Trade name	Chemical name	Chemical Symbol	CAS Number
Toxic	☐ YES ☐ NO				
Harmful	☐ YES ☐ NO				
Corrosive	☐ YES ☐ NO				
Reactive	☐ YES ☐ NO				
Flammable	☐ YES ☐ NO				
Explosive (*)	☐ YES ☐ NO				
Radioactive (*)	☐ YES ☐ NO				
Biological (*)	☐ YES ☐ NO				
Oxidizing	☐ YES ☐ NO				
Sensitizer	☐ YES ☐ NO				
Other dangerous substances	☐ YES ☐ NO				

Goods preparation	
If you have replied YES to one of the above questions. Has the product been purged? ☐ YES ☐ NO	
If yes, which cleaning agent/method:	
Has the product been drained from oil? ☐ YES ☐ NOT APPLICABLE	
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